

git Workshop

Apr 05, 2025

ReDI School Of Digital Integration
Aarhus

Agenda

- 10:00 - 10:30: Theoretical introduction to git
- 10:30 - 11:00: Practical part 1 - Using git in local environment
- 11:00 - 11:15: Water break
- 11:15 - 12:15: Continuation of part 1
- 12:15 - 12:45: Lunch
- 12:45 - 13:30: Practical part 2 - Collaboration
- 13:30 - 13:45: Water break 2
- 13:45 - 14:45: Continuation of part 2
- 14:45 - 15:00: Self learning recommendation
- 15:00 - Thanks for today!

Git

“Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency.”
(git-scm.com)

Version control

- A system that records changes to a file or a set of files over time.
 - It allows you to recall specific versions of those files later.
 - It provides a history of modifications, not just the latest state.
 - It helps manage and track iterations of a project.
-
- **It's essentially a database of changes.**

Why use git?

- Version History,

Why use git?

- Version History,
- Collaboration,

Why use git?

- Version History,
- Collaboration,
- Branching,

Why use git?

- Version History,
- Collaboration,
- Branching,
- Rollback,

Why use git?

- Version History,
- Collaboration,
- Branching,
- Rollback,
- Backup,

Why use git?

- Version History,
- Collaboration,
- Branching,
- Rollback,
- Backup,
- Transparency,

Why use git?

- Version History,
- Collaboration,
- Branching,
- Rollback,
- Backup,
- Transparency,
- Code Reviews,

Why use git?

- Version History,
- Collaboration,
- Branching,
- Rollback,
- Backup,
- Transparency,
- Code Reviews,
- Remote Work

Git Lingo

Staging

The staging area is where you prepare your changes for a commit. It's like a checklist, allowing you to select which modifications will be included in the next snapshot of your work.

Commit

A commit records a snapshot of your staged changes, essentially saving the current state of your project's files. This creates a specific point in your project's version history.

Branch

A branch is a separate line of development, allowing you to work on features or fixes in isolation. It enables parallel work and keeps changes separate until they're ready to be merged.

Merge

Merge integrates changes from different branches, combining code into a single branch. It handles the integration of work done in parallel.

Merge conflict

A merge conflict occurs when Git cannot automatically combine changes from different branches. This happens when multiple people modify the same lines in a file. It requires manual intervention to resolve the conflicting changes.

Repository

A repo, or repository, stores all your project's files and their history, acting as the central location for version control. It can be either local on your computer or remote on a server.

Pull/Push

Pull retrieves changes from a remote repository to your local copy, while push sends your local changes to the remote repository. These commands keep your local and remote repositories synchronized.

Clone

Clone creates a local copy of a remote repository by downloading the project to your computer.